AGSC 2026

School of Post Graduate Studies
Tamil Nadu Agricultural University
Coimbatore – 641 003, Tamil Nadu, India

Proceedings

AGSC 2026

# 11th AGRICULTURAL GRADUATE STUDENTS CONFERENCE

## NEXT-GENERATION AGRICULTURE

CLIMATE ACTION, SUSTAINABILITY AND TECHNOLOGICAL INNOVATION

AUGUST 20th & 21st 2026


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**AUGUST | 20 & 21 | 2026**

*NEXT - GENERATION AGRICULTURE: CLIMATE ACTION, SUSTAINABILITY AND TECHNOLOGY INNOVATION*

*(A NATIONAL LEVEL CONFERENCE FOR THE PG STUDENTS)*

# Abstracts

AGSC 2026


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**BOOK TITLE**

**Next-Generation Agriculture: Climate Action, Sustainability and Technological Innovation**

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## Foreword



Agriculture is witnessing a remarkable wave of innovation, with science, technology and human ingenuity coming together to address soil health, climate challenges and productivity. Technologies such as AI, drones, robotics, precision farming, biotechnology and gene editing are transforming agricultural practices, strengthening climate resilience and promoting efficient use of resources. However, the true potential of these advancements lies in translating cutting-edge research into simple, affordable and impactful solutions for farmers. At this exciting crossroads, meaningful discussions, exchange of ideas and collaborative efforts are essential to shape an agriculture that is smarter, more resilient and sustainable.

With this vision in mind, the 11th Agricultural Graduate Students Conference (AGSC 2026), organized by the School of Postgraduate Studies, Tamil Nadu Agricultural University (TNAU), Coimbatore, provides a vibrant platform for the student fraternity to showcase innovative research centred on "Next-Generation Agriculture: Climate Action, Sustainability and Technological Innovation." Bringing together more than 1,311 participants and 545 research papers, the conference facilitates research sharing, collaboration and innovative solutions to pressing agricultural challenges. Through insightful discussions and collaborative efforts, AGSC 2026 seeks to respond to the demands of the present by translating emerging technologies and scientific innovations into practical solutions for a resilient, sustainable and prosperous agricultural future.

In this spirit, I am confident that the 11th Agricultural Graduate Students Conference (AGSC 2026) will inspire young researchers to pursue their scientific aspirations with renewed purpose and dedication. I sincerely appreciate, Dr. C. Babu, Dean, School of Postgraduate Studies (SPGS), and the faculty members of the SPGS office for their encouragement and support in providing postgraduate students with a platform to present their research aligned with the conference theme. I also congratulate all the postgraduate students for their valuable contributions and research presentations. I hope AGSC 2026 stands as a reflection of the collective commitment of the organizers and participants and marks a meaningful milestone in advancing research and innovation in agricultural sciences.

**Acting Vice-Chancellor**

SCHOOL OF POST GRADUATE STUDIES
Tamil Nadu Agricultural University
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Dr. C. BABU, Ph.D.,
DEAN

# Preface



"Agriculture is the foundation of civilisation and prosperity of every nation" a truth that resonates deeply as we convene for the 11th Agricultural Graduate Students Conference (AGSC), organised by the School of Post Graduate Studies (SPGS), Tamil Nadu Agricultural University, Coimbatore on 20th and 21st August 2026. Over the years, this conference has stood as a beacon for young researchers, giving them a platform to present their work and exchange ideas, bringing the University's diverse research into the spotlight it rightfully deserves, all while addressing the evolving needs of agriculture and society, for a prosperous and empowered nation.

This year's AGSC 2026 is centred on the theme "Next-Generation Agriculture: Climate Action, Sustainability and Technological Innovation." The theme is timely, as agriculture today faces climate change, resource scarcity, and environmental challenges that call for innovative and sustainable solutions to secure its future. In line with this, the conference will unfold across six focused themes: modern approaches to crop improvement and productivity; scientific advances in crop health and protection; sustainable horticultural and forestry practices; efficient post-harvest management and stronger farm-to-market linkages; innovations in agricultural engineering and value addition; and inclusive agricultural innovation that empowers women and rural communities. Together, these themes offer a broad perspective on the opportunities and pathways shaping the future of agriculture.

At this juncture, I sincerely thank our Hon'ble Acting Vice Chancellor, Dr. K. Subrahmaniyan, for his constant support and encouragement in organising the 11th AGSC 2026. I extend my heartfelt appreciation to the Organising Secretaries, Dr. K. N. Ganesan, Dr. T. Selvakumar, Er. G. Vanitha, and Dr. N. Natarajan. My gratitude also extends to the Conference Coordinators, Dr. R. Prabu, Dr. S. Arul Prasad, Dr. R. Abhinayaa, and Mrs. S. Revathi, as well as to all the student secretaries, whose dedication and hard work have made this conference possible. My best wishes to all the participating students, whose enthusiasm and scholarly pursuits are the true driving force behind this event. I am confident that the deliberations during this conference will spark new ideas, inspire meaningful research, and strengthen our efforts to build a more sustainable, resilient, and technology-driven future for agriculture.

Dean (SPGS)

AGSC 2026

# Theme 2

## Scientific Advancements in Nurturing Crop Production



*Abstracts*

AGSC26-02-260

# UAV-BASED CROP COUNT IDENTIFICATION IN A COMMERCIAL POTATO FIELD USING DEEP LEARNING AND GIS

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Potato (*Solanum tuberosum* L.) is a major tuber crop where accurate plant population counts are essential for evaluating emergence, estimating yield potential and guiding in-season management in commercial fields. Manual counting is labour-intensive and impractical at field scale, necessitating an automated image-based approach. A high-resolution RGB orthomosaic of a commercial potato field was acquired using an unmanned aerial vehicle (UAV) and processed as georeferenced GeoTIFF. The field boundary was automatically delineated from the raster's valid-pixel mask, and management sub-blocks were segmented using vegetation-index-based canopy-path separation. The orthomosaic was tiled into 1024 × 1024 pixel overlapping image chips, each retaining its affine transform, tile origin and coordinate reference system to enable accurate geo-referencing of subsequent detections. The tiles were annotated for individual potato plants on the Roboflow platform and used to train a YOLO-based object detection model. At inference, the trained model was applied to all tiles and detections overlapping adjacent tiles were deduplicated using an intersection-over-union (IoU) threshold of 0.5 to avoid double counting plants near tile boundaries. Pixel-space bounding boxes were transformed into geographic coordinates using each tile's affine transform, generating point and bounding-box GIS layers with confidence scores. Detected plants were spatially joined to management-block polygons to compute plant count and density per block, and results were exported as GIS-ready layers, reports and an annotated georeferenced raster. This geospatial deep learning pipeline enabled accurate, scalable and spatially precise plant counting directly usable in QGIS, offering a practical tool for precision monitoring in commercial potato production.

**Keywords:** UAVs, YOLO object detection, Geospatial deep learning, Plant population counting, Precision agriculture.



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